

Numerical Modeling of Flow Dynamics in The Aluminum Smelting Process: Comparison Between Air–Water and CO₂–Cryolite Systems



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Air–water models have been widely applied as substitutes for CO₂–cryolite systems in the study of the complex bubble dynamics and bubble-driven flow that occurs in the molten electrolyte phase in the aluminum electrolytic process, but the detailed difference between the two systems has not been studied. This paper makes a numerical comparison between the bubble dynamics for the two systems. Simulations of both single bubble and continuous bubbling were conducted using a three-dimensional computational fluid dynamics (3D CFD) modeling approach with a volume of fluid (VOF) method to capture the phase interfaces. In the single bubble simulations, it was found that bubbles sliding under an anode in a CO₂–cryolite system have a smaller bubble thickness and a higher sliding velocity than those in the air–water system for bubbles of the same volume. Dimensionless analysis and numerical simulation show that contact angle is the dominant factor producing these differences; the effects of kinematic viscosity, surface tension, and density are very small. In the continuous bubbling simulations, the continuous stream of air bubbles detaches from the anode sidewall after a period of climbing, just as it does in the single bubble simulation, but bubbles have less tendency to migrate away from the wall. Quasi-stable state flow characteristics, *i.e.*, time-averaged bath flow pattern, turbulence kinetic energy, turbulence dissipation rate, and gas volume fraction, show a remarkable agreement between the two systems in terms of distribution and magnitude. From the current numerical comparisons, it is believed that the air–water model is a close substitutive model for studying bubble-driven bath flow in aluminum smelting processes. However, because of the difference in bubble morphologies between the two systems, and also the reactive generation and growth of bubbles in the real system, there will likely be some differences in bubble coverage of the anode in the anode–cathode gap.

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I. INTRODUCTION

THE technology and design of the Hall–Héroult reduction cell has progressed substantially since its invention,^[1] with cell capacity increasing from 4000 to 600,000 A and current efficiency increasing from 75 to 96 pct; however, it is still the only process for primary aluminum smelting at the commercial scale. It is also one of the most energy intensive unit processes in modern industry, requiring about 13,000 kWh electrical energy to produce one ton of primary aluminum metal, of which only approximately 50 pct of the energy is

required for reaction, with the other half generating waste heat which is emitted to the atmosphere.

In a modern Hall–Héroult reduction cell, illustrated in the schematic diagram in Figure 1, direct current is fed into dozens of suspended anodes, passes through an electrolyte layer and metal pad to the cathode which lies at the base of the cell, and finally flows out through collector bars to next cell. Alumina (Al₂O₃) is added periodically into the molten bath [at approximately 1223 K (950 °C)], from several feeders. Under the effect of direct current, the aluminum (Al) metal is reduced from alumina and deposited at the cathode and is intermittently pumped out; while oxygen (O) from the alumina combines with carbon (C) in the anode to produce carbon dioxide (CO₂) at the bottom surface of the anode.

The CO₂ gas forms as dispersed bubbles in the molten salt which slide along the bottom surface of the anode and release at the anode edges where they then rise in channels around the anodes, before finally releasing into the atmosphere. These bubbles play a complex role in cells, not only for bath circulation, which affects alumina mixing and heat exchange, but also in affecting the cell resistance and current distribution. Thus, a

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better understanding of the bubble dynamics and the resulting liquid flow are keys to improve cell performance.

Due to the corrosive environment, it is difficult to study the bubbles and the resultant flow directly in a real cell. Various model substitutes have been developed, including water models,^[2–23] low temperature electrolytic models,^[24–28] small-scale high-temperature electrolytic cells,^[29–43] and numerical or computational fluid dynamics (CFD) models.^[4,17,18,22,44–64] With their easier operation and measurement and the possibility of full scale, water models have been used widely. In addition to provide a better understanding of the process, water models have often been used to validate CFD models, either based on flow patterns^[4] or detailed liquid velocity and turbulence^[51–53] obtained using Particle Image Velocimetry (PIV) technology.^[15] The validated CFD model is then extended to simulate the exact scale, properties, and operating parameters of the real system.^[54] Apart from the ready availability and inert qualities of the working fluids, the reason for using an air–water system to represent the CO₂–cryolite system is that the kinematic viscosity of water is very similar to that of cryolite [$1.43 \times 10^{-6} \text{ m}^2/\text{s}$ for cryolite at 1223 K (950 °C) and $1.005 \times 10^{-6} \text{ m}^2/\text{s}$ for water at 298 K (25 °C)]. With the similar kinematic viscosity, it is believed that similar liquid flow dynamics will result as long as the same volume of gas bubbles is used. However, there are some undetermined factors: the other gas and liquid properties (such as surface tension, contact angle, and density) are quite different in the two systems. In fact, there is good evidence that bubble shape and motion are strongly influenced by the liquid properties.^[65,66] Consequently, in order to have confidence in using air–water models to simulate

CO₂–cryolite systems, it is essential to quantitatively identify similarities and differences in the bubble dynamics of the two systems.

Experimental comparison between the bubbles dynamics for the two systems is difficult: experimental measurements of bubble dynamics cannot be made in real cells. Small-scale high-temperature transparent electrolytic cells have been used to study bubble dynamics in a real electrolytic environment, but due to size limitations, bubble flow is not fully representative of that occurring in actual electrolytic cells. Furthermore, it is more difficult to assess the dominant factor on the detailed difference in bubble dynamics. For instance, to evaluate the effect of the viscosity on bubble motion experimentally, two liquids are needed that have similar fluid properties except for viscosity.

In a recent work,^[56] a two-dimensional (2D) CFD-VOF model has been used to numerically analyze similarities and differences in bubble behavior of air–water and CO₂–cryolite systems. It was found that both systems show similar trends in bubble dynamics. The difference is that the air bubble presents a larger thickness and a smaller sliding velocity under the anode, and detaches from the side wall while rising. The detachment involves the loss of three-phase (air–water–anode) contact after which the bubbles migrate away from the wall during their subsequent rise. On the other hand, the CO₂ bubbles remain attached to the side wall for their entire rise to the surface. Subsequently, a further investigation was carried out to address the dominant factors affecting the difference in bubble behavior in the two systems.^[57] The results showed that contact angle plays a dominant role. However, there are two remaining uncertainties in these assessments.^[56,57] First, these investigations were conducted in 2D, and so

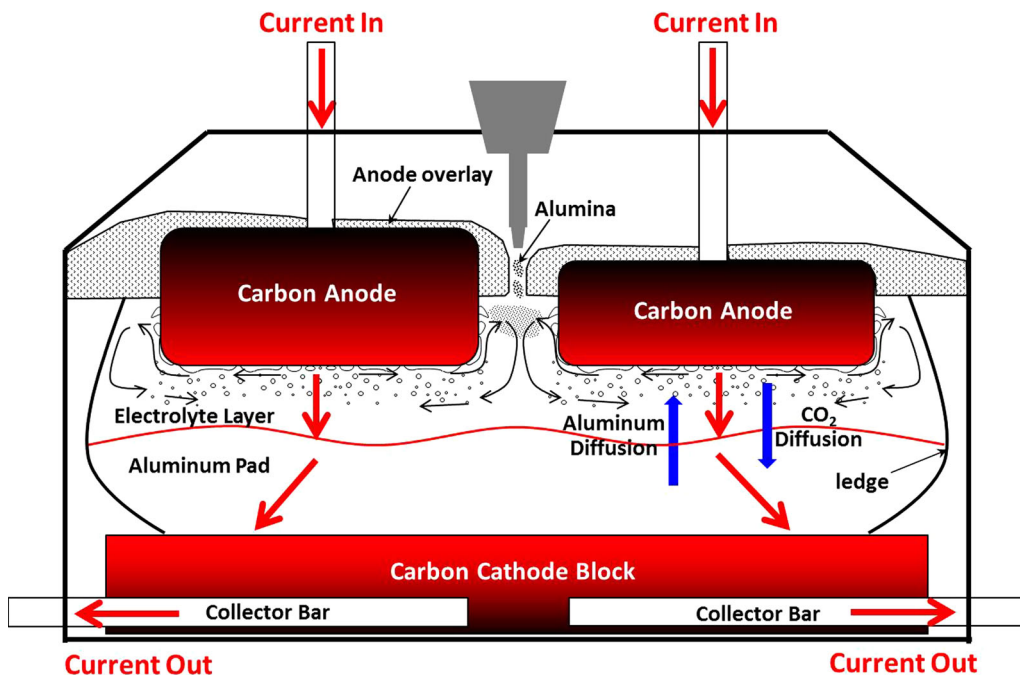


Fig. 1—Schematic diagram of a typical Hall–Héroult reduction cell.

do not represent the actual three-dimensional (3D) bubble shape. As the bubble is in reality generated in an intrinsically 3D arrangement, it is important to know whether a 3D simulation would give the same result. Second, the investigations were for a single isolated bubble. In reality, the bubbles are generated in a continuous manner in reduction cells, so it is necessary to address how much difference exists between the two systems with continuous bubble generation.

In this work, a 3D CFD-VOF model has been used to predict 3D bubble shape and dynamics. Following model validation, the motion of a fixed size bubble was simulated, including the stages of bubble generation, sliding under the anode and rising in a side channel. The model

was used to compare the two physical systems, and also determine the dominant factor affecting the difference in bubble dynamics. The model was then extended to run with continuous bubble generation for both systems, and a detailed comparison was carried out in terms of time-averaged results, *e.g.*, the flow pattern, turbulence kinetic energy, turbulence dissipation rate, and gas volume fraction distribution. Furthermore, these time-averaged results can be used to compare with results from time-averaged bath flow models and to build constitutive correlations to improve macro-scale modeling accuracy if necessary. This multi-scale modeling approach has received increasing interest in the study of complex multiphase flow systems.^[64,67–70]

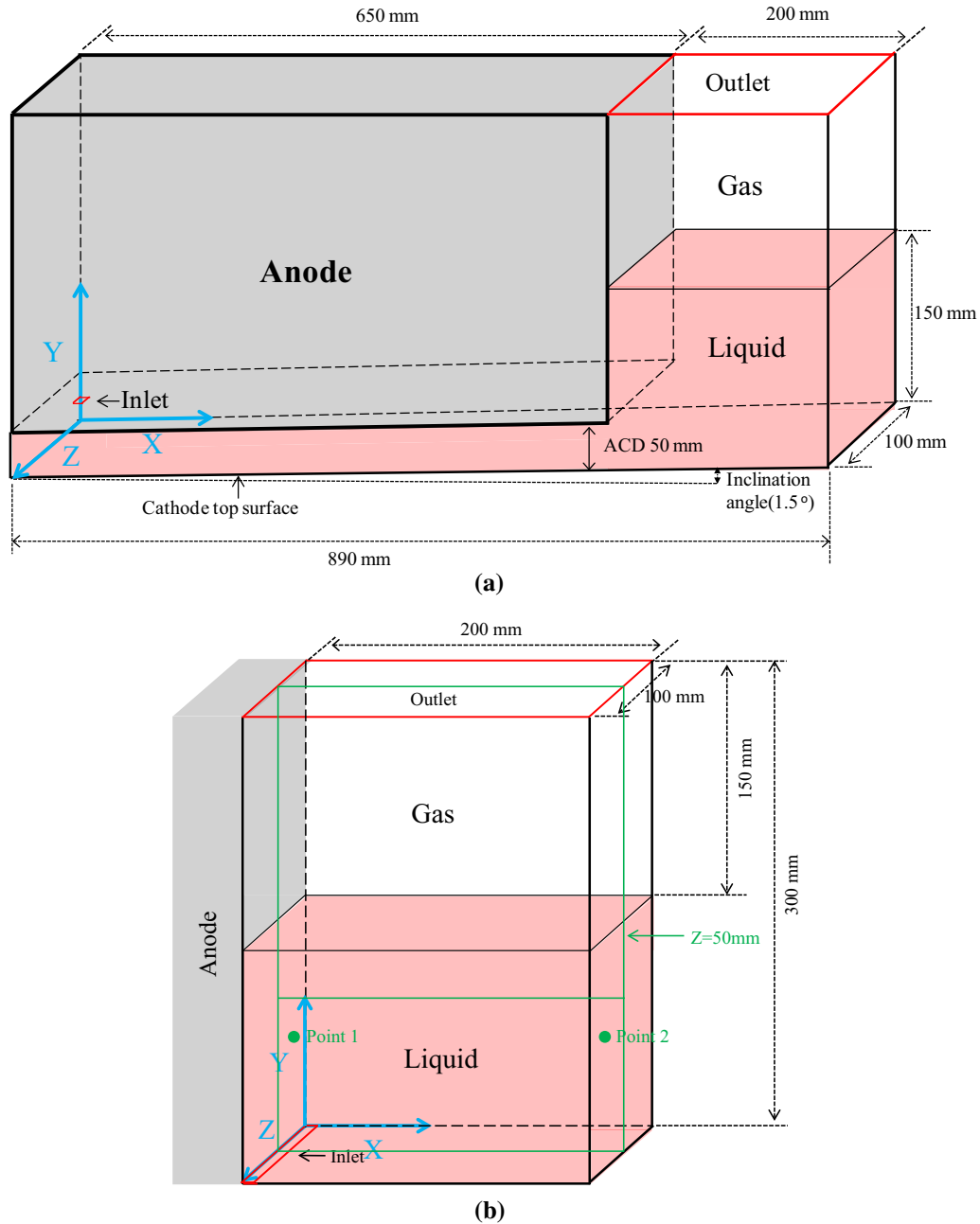


Fig. 2—Geometries of the simulation cases: (a) Geometry 1 for single bubble simulation; (b) Geometry 2 for continuous bubbling simulation.

Table I. Cases Conducted in this Paper

Cases	Systems	Contact Angle (deg)	Geometry
Case 1	CO ₂ -cryolite	60	geometry 1
Case 2	air-water	120	
Case 3	air-water	60	
Case 4	CO ₂ -cryolite	60	geometry 2
Case 5	air-water	120	

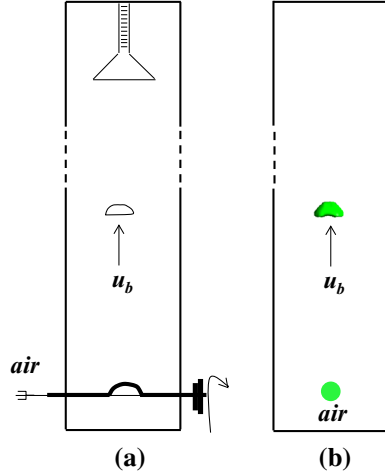


Fig. 3—Model validation using single bubble rising data: (a) experiment; (b) CFD model.

II. MODEL DESCRIPTION

In this paper, the VOF method was applied using the CFD software ANSYS Fluent 15.0. The details of this method and the key governing equations are well documented in the ANSYS Fluent User Manual,^[71] and they are not repeated here.

As the real bubble behavior in commercial cells is coupled with complex phenomena including thermal transport, alumina transport, chemical reaction and current distribution, some simplifications are made in this study so as to focus only on the bubble flow dynamics:

- the gas and liquid phases are incompressible,
- the two-phase flow is isothermal,
- there is no chemical reaction in the domain,
- the effect of electromagnetic force is ignored.

Two geometries were built in this paper for different purposes, as shown in Figure 2. Geometry 1 represents a slice through a typical commercial Hall-Héroult cell. The dimensions of the geometry are based on the previous 2D simulations^[56,57]; the only difference being that the thickness of the slice is now set to 100 mm. This geometry is used to simulate single bubble dynamics, with the aim of assessing similarities and differences between the two systems so that the dominant factors affecting these differences can be determined. As listed in Table I, three cases are carried out using this geometry: cases 1 and 2 represent the real CO₂-cryolite system and the air-water system, respectively, and case 3 is a system

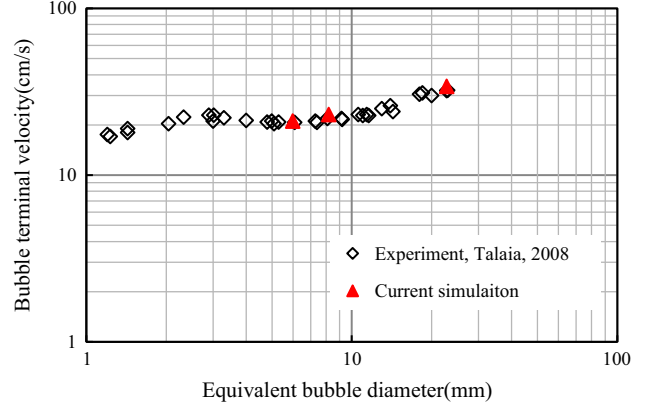


Fig. 4—Comparison between experimental and simulation rise velocities of different sized bubbles.

in which the liquid and gas properties are those of water and air, except for contact angle which is taken to be 60 deg in order to isolate the effect of this parameter.

Geometry 2, shown in Figure 2(b), is used to simulate continuous bubbling. The movement of bubbles under the anode is omitted in these simulations: it would be very time consuming to simulate a continuous stream of many bubbles under an anode, so the process is simplified to only account for the period after bubbles escape from the ACD. This simplification should be acceptable, given that the main driver for liquid flow in the side channel is the bubble motion in the channel. Two cases apply using this geometry, as listed in Table I: case 4 represents the real CO₂-cryolite system and case 5, an air-water system.

As a boundary condition, gas is injected into the liquid to represent gas generation. This is indeed the mechanism of bubble generation in the air-water system but in a real CO₂-cryolite system gas is generated by the electro-chemical reaction. In this work, we focus our attention on bubble dynamics after generation, so this gas generation setting is acceptable.

In geometry 1, a uniform normal velocity (0.25 m/s) is applied at the inlet boundary (5 mm × 5 mm), which is 20 mm from the left wall. This inlet condition is only applied to form the initial bubble with a fixed diameter. After injecting for 0.12 second, the inlet boundary reverts to a non-slip wall boundary. For the purpose of comparing the present 3D simulations and the previous 2D simulations^[56], the equivalent diameter in the 2D simulations^[56] was applied in our 3D simulations. The difference is the equivalent diameters of 3D bubble and 2D bubble are based on the ratio of gas

volume to surface area (3D) and the ratio of surface area to perimeter (2D), respectively. The anode-cathode distance (ACD) is 50 mm. The initial liquid depth is taken to be 200 mm, as is typical of operating cells.

In geometry 2, we assumed that the gas generated at the anode bottom is 100 pct CO₂. Based on Faraday's law, one mole of CO₂ will be produced for four moles electrons passing through the circuit; the gas generation rate per unit area due to current densities can be expressed as follows:

$$q = \frac{i \times R \times T}{4 \times F \times P}, \quad [1]$$

where q is the gas generation rate per unit area of anode bottom surface, m³/(m² s); i is the anodic current density, 0.7 A/cm² here; R is the universal gas constant, 8.314 J/(mol K); T is the absolute temperature, 1223 K (950 °C); F is Faraday's constant, 96,485 A/(mol s); and P is atmospheric pressure, 101,325 Pa.

If the gas release is assumed to be evenly distributed along four edges of the anode, the gas injection velocity at the inlet boundary can be calculated as:

$$V = (q \times a \times b) \times \frac{l}{2a + 2b} \times \frac{1}{l \times w}, \quad [2]$$

where V is the gas injection velocity at the inlet boundary, m/s; a is the length of anode bottom surface, 1300 mm; b is the width of anode bottom surface, 650 mm; l is the length of inlet boundary, 100 mm; and w is the width of inlet boundary, 5 mm. It should be emphasized that the anode dimensions are not related to any specific anode design, but are typical.

For both geometries, the top surface opening to air is taken to be an outlet boundary condition with a constant atmospheric pressure. The flow direction is normal to the surface, with any backflow being 100 pct gas. A no-slip condition is applied for both phases at the boundaries of all walls.

To deal with the surface tension at the interface, the continuum surface force model (CSF)^[72] is applied, in which the contact angle is incorporated in the wall adhesion. The values of surface tension are 0.132 N/m for cryolite at 1223 K (950 °C) and 0.072 N/m for water at 298 K (25 °C). Mesh adaption technology is applied here. The grid on the gas-liquid interface is refined and coarsened as bubbles pass, which significantly reduces the computing time.

Regarding the numerical solution procedure, the pressure-implicit method with splitting of operators (PISO) is used as pressure-velocity coupling; pressure staggering option (PRESTO!) is applied for pressure discretization; and the quadratic upstream interpolation for convective kinematics (QUICK) is applied for

momentum discretization. The absolute convergence criteria for continuity and velocity were both set to be 10⁻⁷.

III. RESULTS AND DISCUSSION

A. Model Validation

In order to validate the numerical model, it is sufficient to use experimental data (bubble terminal rising velocity and bubble shape) available from the literature^[73] for this purpose. The reported experiment was carried out in a tank of size 200 mm × 200 mm × 1500 mm as shown in Figure 3(a). The gas was introduced and trapped inside an inverted cup; when it rotated, a bubble was produced near the base of the tank and rose freely in water. The experimental velocity can be calculated by the distance traveled in every time interval. The terminal velocities of various sized bubbles were plotted and are shown in Figure 4. It is not necessary to validate all bubbles in Talaia's work here; only three bubbles were chosen as examples: air bubbles with diameters of 6, 8.2, and 22.8 mm. The actual experimental geometry, as shown in Figure 3(b), was simulated. The tank was full of water and a spherical bubble was initially patched near the base. The bubble rose freely in the distilled water when the computation was started. With the above model settings, the bubble terminal rising velocities determined for the above three cases are plotted in Figure 4 together with the experimental values. The model predictions show good agreement with the experimental data, indicating that the present settings of the VOF model are reliable.

B. Single Bubble Model

1. Dimensionless analysis

For a gas bubble in a liquid that is moving along the surface of an anode, a function dependent on various system variables that describes the bubble morphology and motion can be proposed:

$$F_{\text{bubble}} = f(\rho_l, \rho_g, \Delta\rho, \mu_l, \sigma_l, d_g, v_g, g, \theta, X), \quad [3]$$

where ρ_l , μ_l , and σ_l are the liquid density, dynamic viscosity, and surface tension respectively; v_g and d_g are the bubble velocity and equivalent diameter; $\Delta\rho$ represents the density difference between the liquid and gas; g is the gravitational acceleration, 9.81 m/s²; θ is the gas-liquid contact angle on the anode surface; and X stands for all other factors, *e.g.*, surface heterogeneity, surface roughness, and anode inclination.

Based on the Buckingham π theorem, the original six variables (ρ_l , ρ_g , μ_l , σ_l , v_g , and d_g) and the density

Table II. The Controlling Parameters for the Re Number and We Number in the Two Systems

Controlling Parameters	Cryolite at 1223 K (950 °C)	Water at 298 K (25 °C)
μ_l/ρ_l for Re number	1.43×10^{-6}	1.005×10^{-6}
σ_l/ρ_l for We number	6.29×10^{-5}	7.21×10^{-5}

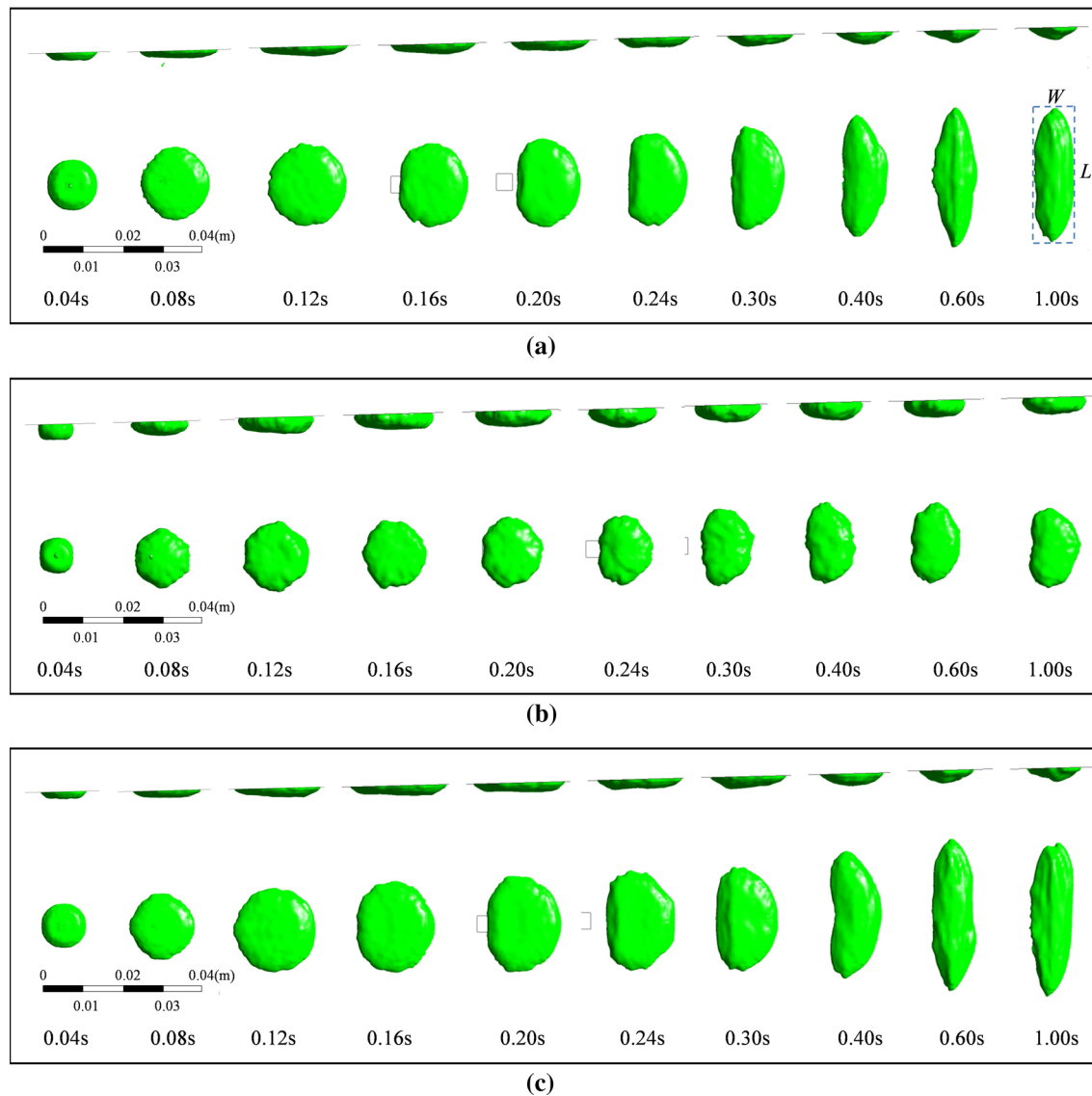


Fig. 5—Evolution of bubble morphologies in ACD at different time instants. (a) Case 1, CO₂-cryolite, contact angle 60 deg; (b) Case 2, air-water, contact angle 120 deg; (c) Case 3, air-water, contact angle 60 deg.

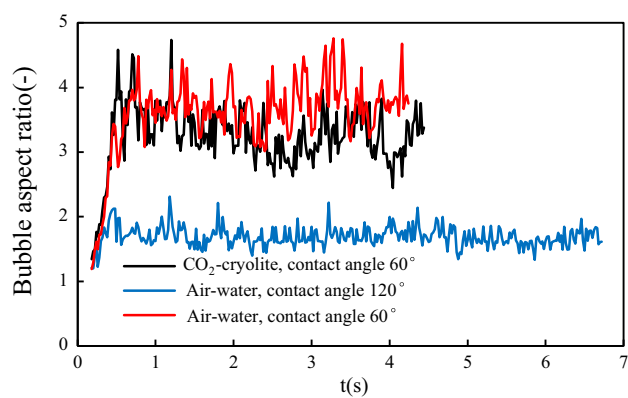


Fig. 6—Variations in bubble aspect ratio with time in ACD for three cases.

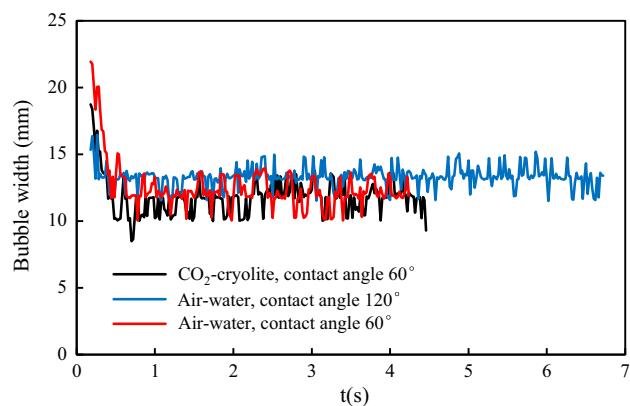


Fig. 7—Variations in bubble width with time in ACD for three cases.

difference ($\Delta\rho = \rho_1 - \rho_g$) in Eq. [3] can be rewritten in terms of three dimensionless parameters π_1 , π_2 , and π_3 .

$$\pi_1 = \frac{\rho_1 v_g d_g}{\mu_l} = \frac{v_g d_g}{\mu_l / \rho_1},$$

$$\pi_2 = \frac{\rho_1 v_g^2 d_g}{\sigma_l} = \frac{v_g^2 d_g}{\sigma_l / \rho_1},$$

$$\pi_3 = \frac{\rho_1 v_g^2}{\Delta\rho g d_g} = \frac{v_g^2 / g d_g}{\Delta\rho / \rho_1},$$

where π_1 , π_2 , and π_3 represent three important dimensionless numbers: Reynolds number (Re), Weber

number (We), and Froude number (Fr), respectively. Consequently, Eq. [3] can be written as follows:

$$F_{\text{bubble}} = F(Re, We, Fr, \theta, X). \quad [4]$$

For a fixed bubble size, it is surprising to notice that the controlling parameters for the Re number and We number are very close in value, as shown in Table II. This indicates the dynamic viscosity and surface tension differences cause almost no difference in the bubble behavior of air–water and CO_2 –cryolite systems.

2. Numerical analysis

In this section, we will compare results from numerical simulations to investigate the effects of density difference and the contact angle on bubble dynamics. By quantifying the bubble morphologies, aspect ratios, and bubble velocities, case 1 and case 3 are compared to investigate the effect of density difference, and case 2 and case 3 are compared to focus on the effect of contact angle.

a. Bubble generation and sliding beneath anode in ACD. Figure 5 shows the morphologies of bubble growth and sliding beneath the anode in three cases at different time instants. Both top view and side view are shown together to clearly illustrate the 3D bubble evolution.

Initially, a single bubble grows at the nucleation point due to gas injection. The bubble attaches to the bottom surface of the anode. The bubble grows until it reaches a maximum thickness due to it being squeezed flat by the buoyancy force, thus forming a ‘pancake’-shaped bubble. With further gas injection, the bubble expands along the anode surface.

When the bubble grows sufficiently large for buoyancy to overcome the surface forces holding it in place, it begins to slide along the anode bottom surface. In our simulation, we close the inlet boundary after a fixed time of 0.12 second to quantitatively investigate a fixed bubble size. The bubble moves from left to right along the tilted anode bottom surface. Several bubble shapes in the evolution process are identified: a nearly perfect sphere, a semi-ellipsoid, and finally an elongated bubble. All of the bubble shapes were observed in the experiments.^[5,13] In the sliding process, the bubble suffers considerable resistance from the liquid in front, which squeezes it to an elongated shape. For some conditions, the bubble presents a larger thickness in front and a small tail at the rear part of the bubble: a shape that has been called a ‘Fortin bubble’.^[3] In our previous 2D work,^[56] the bubbles with the same equivalent diameter presented a ‘Fortin’ shape. We cannot see the ‘Fortin’

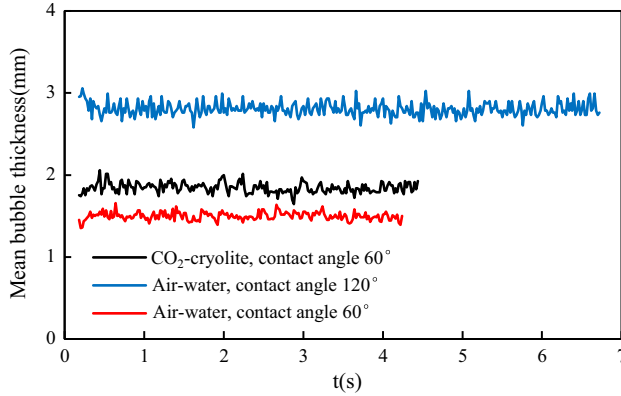


Fig. 8—Variation in mean bubble thickness with time in ACD for three cases.

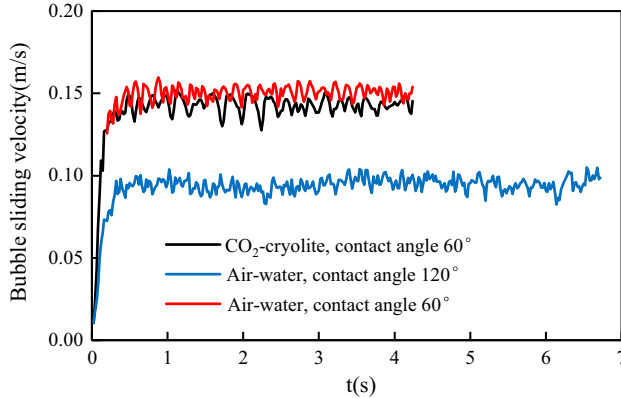


Fig. 9—Variations of bubble sliding velocities with time in ACD for three cases.

Table III. Comparisons of Present 3D Simulation with 2D Simulation and Experimental Studies

	Diameter (mm)	Mean Sliding Velocity (m/s)
Experiment, air–water system ^[12]	10.0	0.08
Experiment, air–water system ^[12]	11.3	0.10
2D simulation, air–water system ^[56]	11.3	0.09
3D simulation, air–water system	11.3	0.10

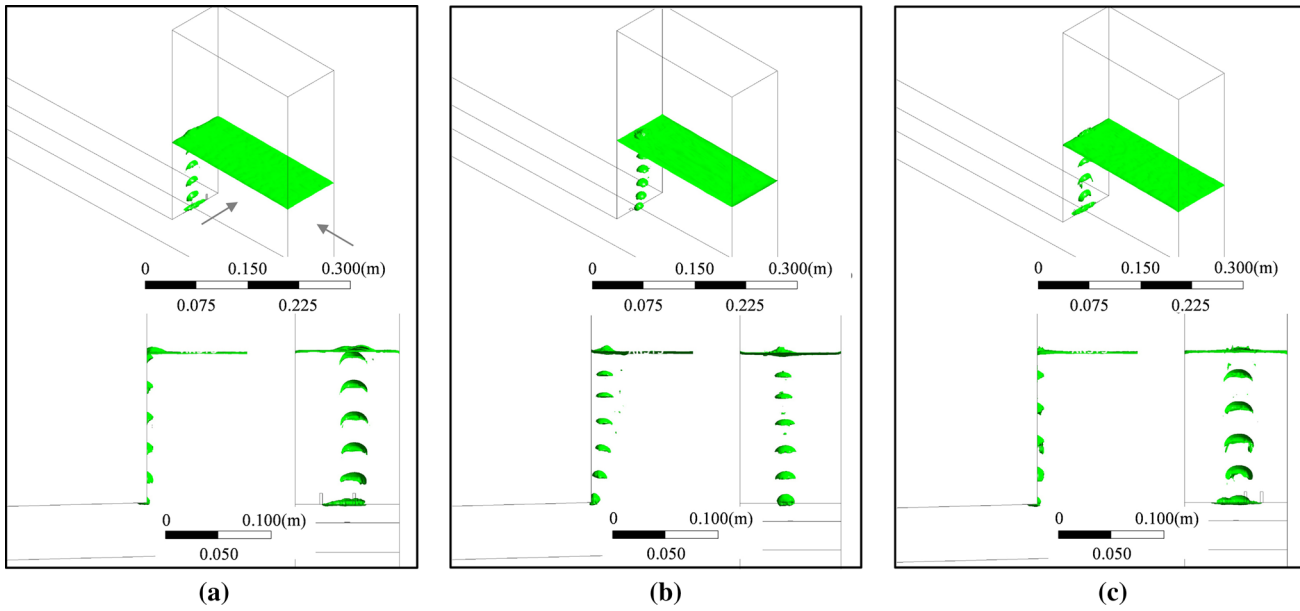


Fig. 10—Bubbles rising in the side channel. (a) Case 1, CO₂-cryolite, contact angle 60 deg, $t = 4.5, 4.6, 4.7, 4.8, 4.9, 5.0$ s; (b) Case 2, air-water, contact angle 120 deg, $t = 6.8, 6.9, 7.0, 7.1, 7.2, 7.3, 7.4$ s; (c) Case 3, air-water, contact angle 60 deg, $t = 4.3, 4.4, 4.5, 4.6, 4.7, 4.8$ s.

shape in the current simulation. It is likely that the small bubble size contributes to this phenomenon. This indicates that 2D simulation cannot fully represent the actual 3D case, and it is necessary to run the model at 3D to obtain some quantitative results.

There are significant differences in the morphologies between the real CO₂-cryolite system (case 1) and an air-water system (case 2). For the same gas volume, the CO₂ bubbles present a larger covering area and a smaller bubble thickness compared with that of air bubbles. However, if the contact angle is changed from 120 deg (the true contact angle of air-water system) as in case 2, to 60 deg (the contact angle of the CO₂-cryolite system) as in case 3, the bubbles show a much higher degree of similarity with that in case 1 (the real CO₂-cryolite system).

After a period of sliding, the bubble reaches a quasi-stable elongated shape (as shown in Figure 5), which can be expressed by the aspect ratio

$$r(t) = \frac{L(t)}{W(t)}, \quad [5]$$

where L is the length of the sliding bubble, with the axis perpendicular to the direction of the bubble movement, W is the width of the sliding bubble, with the axis parallel to the direction of the bubble movement, as shown in Figure 5(a). Although the aspect ratio cannot describe the bubble shape perfectly, it gives a quantitative measure about the basic tendency of bubble evolution.

Figure 6 shows the instantaneous aspect ratio as a function of time for the 3 cases. Generally, the bubble changes from a nearly perfect circular shape ($r = 1$) into a transversely elongated shape ($r > 1$) during the sliding process. As the bubble begins to move, the aspect ratio climbs to the peak value very quickly, then it fluctuates around an average value. The bubbles in case

1 and case 3 present more elongated shapes in the direction perpendicular to the bubble movement. From Figure 6, we can see there is sufficient time to sample the time-averaged aspect ratio. The mean value of aspect ratio is averaged after 1 second. For case 1 and case 3, the average values are 3.6 and 3.8 respectively, and for case 2, it is 1.7. Also the higher magnitude of the fluctuations in case 1 and case 3 indicates a stronger bubble deformation than in case 2.

Figure 7 illustrates the variation in bubble width with sliding time, which gives additional information on the bubble evolution. It can be seen that all three cases present a similar trend with the bubble width dropping to a quasi-stable value after 1 second and then fluctuating around the mean value. However, the bubble width is very similar in all three cases, although the case with contact angle 120 deg has a slightly larger width. This indicates that bubble width is more critical than aspect ratio. In fact, the previous 2D simulations^[56] effectively assumed infinitely long bubbles. The bubble width formed in such a situation would be a function of anode slope, density, and surface tension, and to some extent contact angle. In a real 3D situation, multi-dimensional effects as well as potential instabilities will limit the length to which bubbles can grow by coalescence.

Figure 8 shows the mean bubble thickness for all cases. For bubbles with the same diameter, the air bubbles are predicted to have a larger thickness compared with CO₂ bubbles, which agrees with previous 2D simulations.^[56] However, the bubble thicknesses in the 3D simulations are 1 mm smaller than those in the 2D simulations, in both air-water and CO₂-cryolite systems. The CO₂ bubble thickness is 1.84 mm in the current 3D simulations, and it is about 3 mm in the 2D simulations. The air bubble thickness is 2.80 mm in the present 3D simulations and is approximately 4 mm in

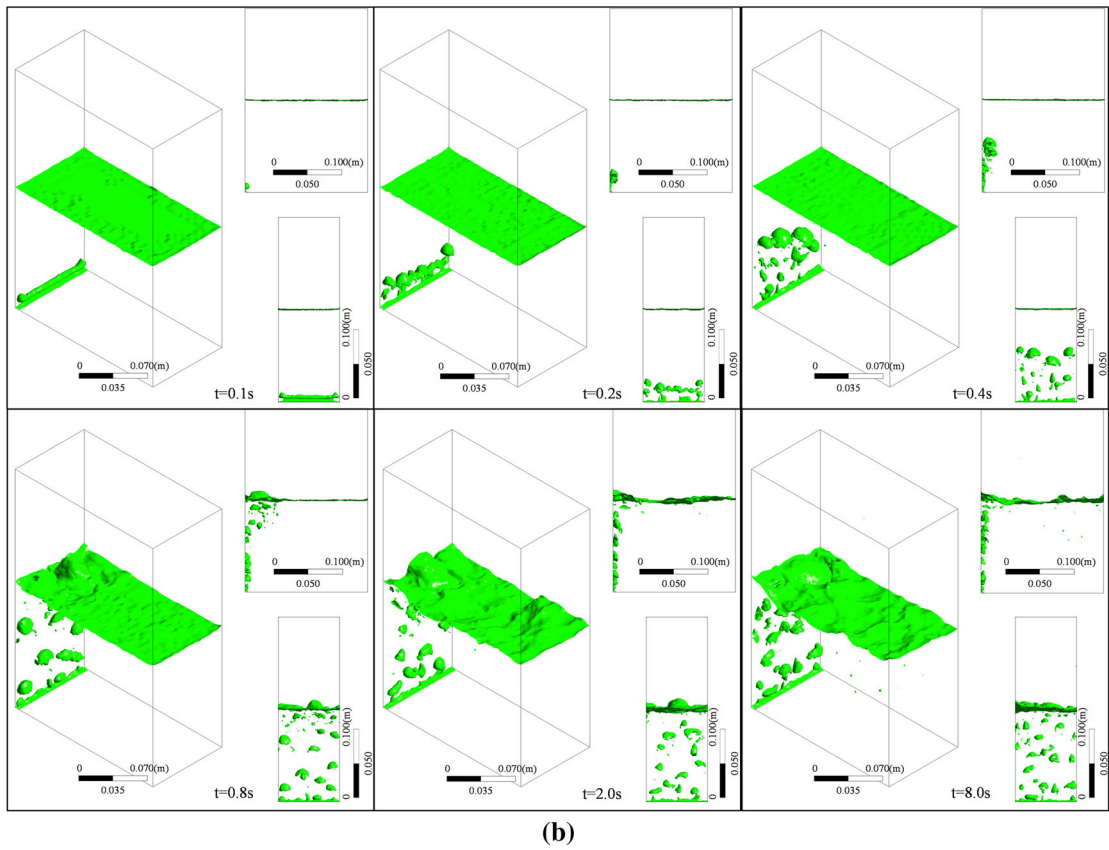
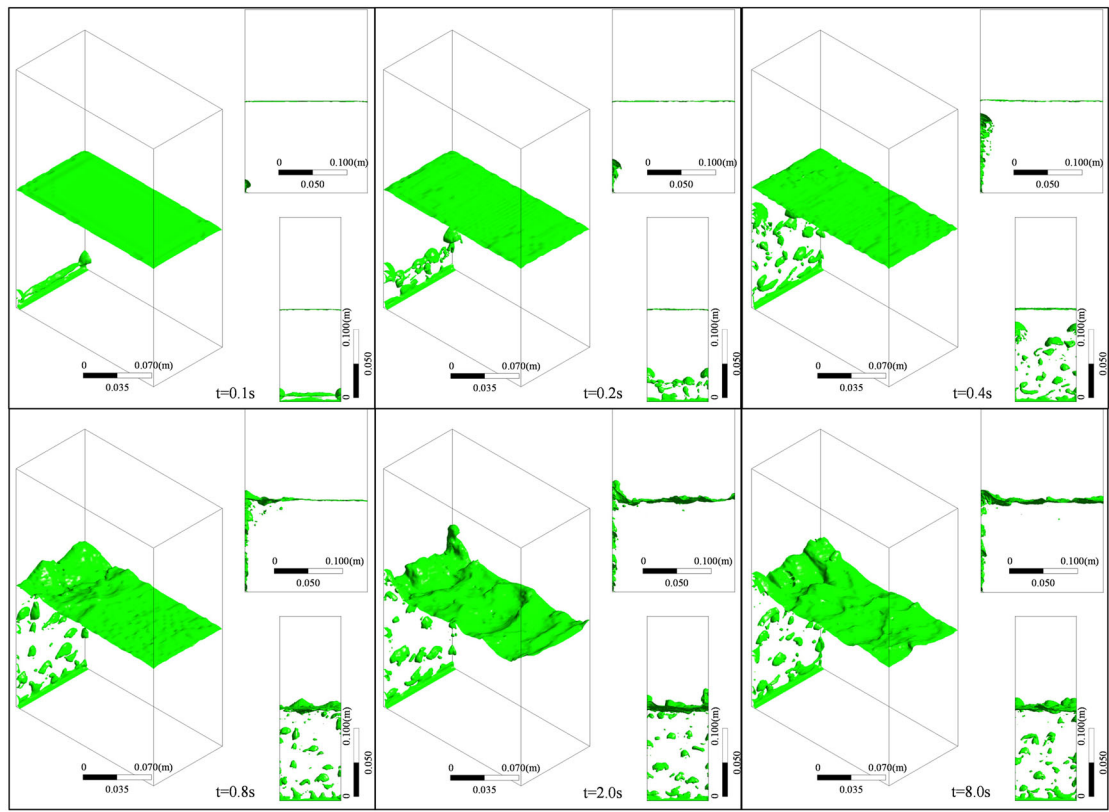


Fig. 11—Bubble morphologies in side channel: (a) Case 4, CO_2 -cryolite system; (b) Case 5, air-water system.

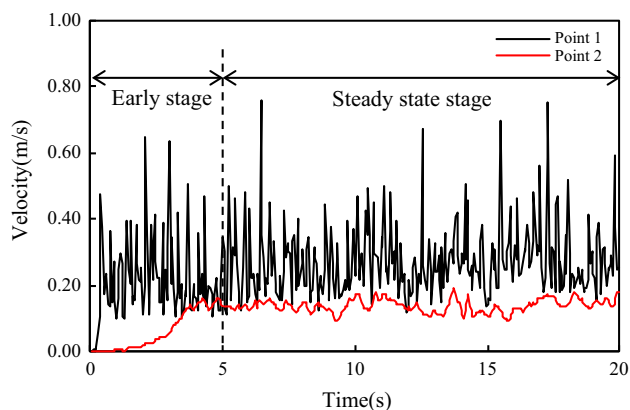


Fig. 12—Instantaneous liquid velocity plotted against time at two points from the transient simulation Case 4: Point 1 (5, 75, 5 mm), Point 2 (195, 75, 5 mm).

the 2D simulations. It also can be clearly seen the effect of contact angle that the bubble thickness in the artificial case 3 is close to the value of CO_2 in cryolite.

The instantaneous bubble velocity is computed from bubble position at each time step and is illustrated in Figure 9. The bubble experiences a very short acceleration period before reaching its steady state regime in less than one second. Like the aspect ratio calculation, the mean value of sliding velocity is averaged after 1 second. The bubble sliding velocity in case 1 (CO_2 -cryolite system with contact angle 60 deg) is consistent with that in case 3 (air-water system with contact angle 60 deg). The time-averaged velocities are 0.14 m/s for case 1 and case 3, and it is 0.10 m/s for case 2. The present 3D simulation of air bubble sliding in water is compared with experimental studies^[12] and with the 2D simulation,^[56] and comparisons are listed in Table III. It is found that in the 2D simulation, the bubble velocity is 10 pct smaller than in the 3D simulation and experimental data. These results show that a 2D simulation does give a qualitative prediction of bubble behavior; however, a 3D simulation gives better agreement with experiment and is necessary for quantitative analysis.

In this section, we have seen that the bubble morphologies, aspect ratio, bubble thickness, and bubble sliding velocity are surprising similar in a CO_2 -cryolite system and an air-water system with contact angle changed to that of the CO_2 -cryolite system. These results indicate that the contact angle plays a dominant role in producing the difference in bubble behavior beneath anodes that is seen when comparing simulations of CO_2 -cryolite and air-water systems.

b. Bubble rising in a side channel. As bubbles are released from the edge of an anode, the bubble movement changes its direction from horizontal to vertical due to the buoyancy effect. Figure 10 plots bubble rising morphologies in the side channel for three cases at different times. For each case, three views are shown: 3D view (top), front view (lower right), and side view (lower left).

Some differences between the two systems can be observed. The CO_2 bubble in case 1 mainly climbs along

the sidewall and remains attached to it, in the sense of maintenance of three-phase (gas-liquid-anode) contact. The air bubble in case 2 detaches from the sidewall after a period of climbing, and three-phase contact is lost; the bubble proceeds to migrate away from the wall as it continues its rise. This behavior is consistent with the higher contact angle for the water case—the surface force holding the bubble to the anode surface is lower than in the case of cryolite. The second difference is that all gases in case 1 transfer from anode bottom to anode side, while the bubble in case 2 breaks off when the bubble releases at the anode edge, leaving a small amount of gas trapped at the anode edge. This result is also consistent with the higher surface force acting in the case of cryolite. As with the bubble morphologies beneath the anode, the bubble behavior in case 3 shows a high degree of agreement with that in case 1.

The mean rise velocity of bubbles is 0.20 m/s for case 1, 0.19 m/s for case 2, and 0.20 m/s for case 3. In general, the rising velocities are not greatly different in all three cases.

In summary, 3D simulations show a similar trend to previous 2D simulation,^[56] e.g., the bubble detaches from the side wall in the air-water system and the bubble remains attached to the side wall in the CO_2 -cryolite system. However, at the same equivalent bubble size, the 3D bubble morphology is quite different from the 2D bubble morphology, and the velocity is also different. This demonstrates that 3D modeling is necessary to predict bubble dynamics quantitatively.

C. Continuous Bubbling Model

In the last section, we used a model of a single isolated bubble to compare the bubble dynamics of the two systems. However, the bubbles are generated in a continuous manner in real aluminum reduction cells, and so in this section, analysis of bubble behavior for continuous injection is carried out.

1. Bubble morphologies

In our simulation, the gas is injected into the liquid from the left bottom of the domain with a fixed velocity. Figure 11 gives some snapshots of the continuous bubbling simulation, using the iso-surface of 0.5 gas fraction to delineate the bubble boundary. Figure 11(a) is the result for CO_2 bubbles rising in cryolite, and Figure 11(b) shows air bubbles rising in water.

The two systems show similar bubble morphology. At the beginning, a horizontal gas tube is formed at time 0.1 second. As gas injection continues, the gas tube breaks up into various sized bubbles due to instability and strong bubble liquid interaction, as shown at time 0.2 second. These bubbles rise in the liquid (at time 0.4 second) and finally break through the liquid surface (at time 0.8 second). The bubbles break and escape at the top liquid surface causing some surface wave motion. The waves tend to be more energetic as time progresses. From these figures, we can see most bubbles concentrated near the anode sidewall in the domain. Although the bubble size is believed to be strongly affected by surface tension, we cannot see clear

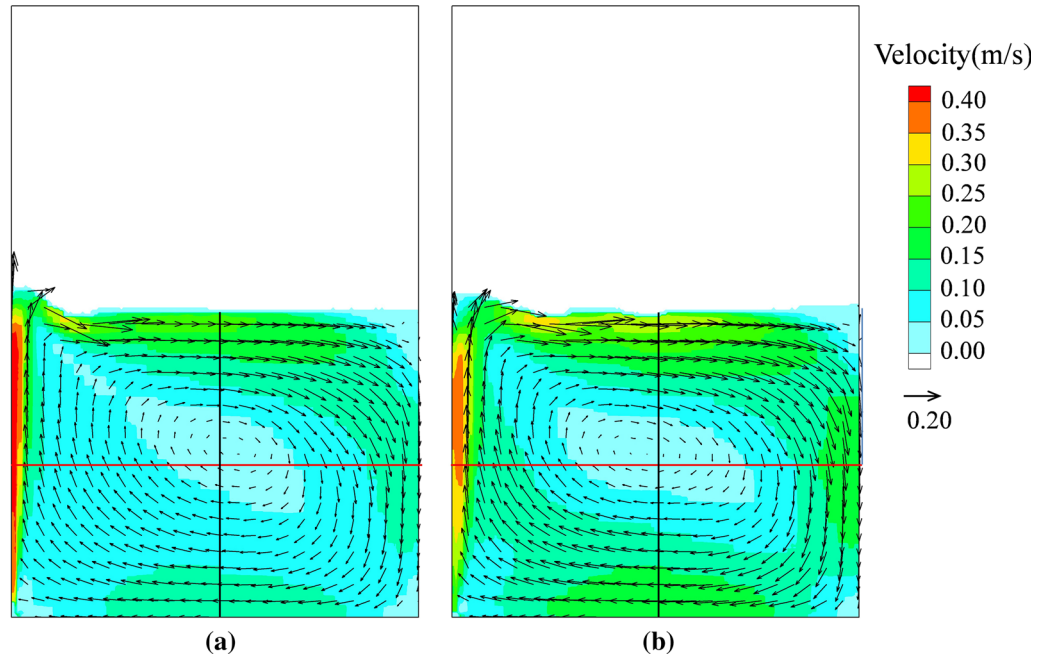


Fig. 13—Time-averaged flow pattern at the middle plane of the simulation domain: (a) Case 4, CO₂-cryolite system; (b) Case 5, air-water system.

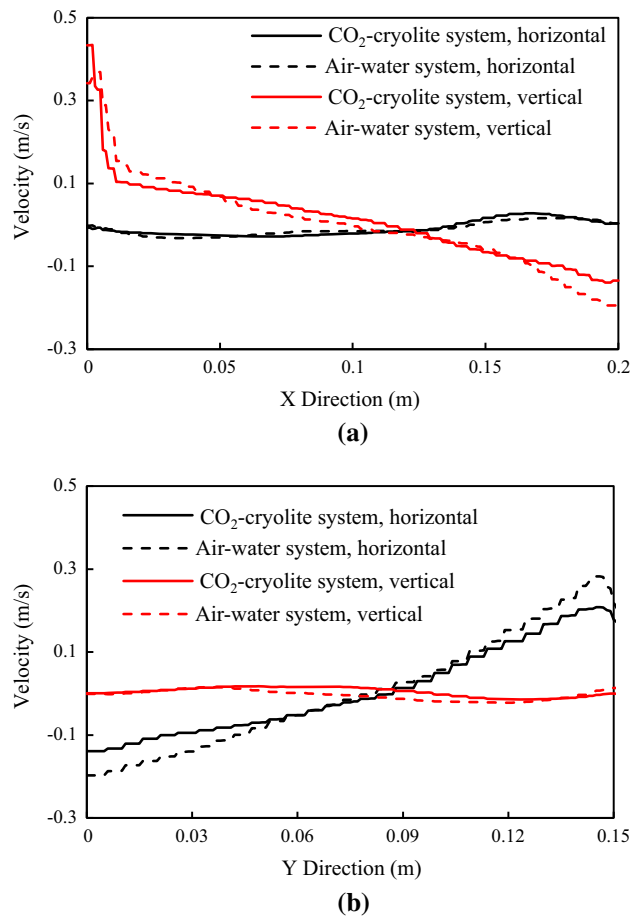


Fig. 14—Time-averaged velocity profiles on two lines for both cases, (a) $Y = 75$ mm, (b) $X = 100$ mm.

differences in bubble size from these pictures. This would be consistent with Weber number being the determining criterion for bubble break-up, as is generally believed. Weber number depends on the kinematic surface tension (σ_1/ρ_1) between gas and liquid which is very similar in the CO₂-cryolite and air-water systems.

Nonetheless some differences between the two cases can be observed. As with the single bubble simulations described in Section III-B-2, CO₂ bubbles remain attached to the anode sidewall as they rise upward, while the air bubbles detach from anode sidewall after a period of climbing (*e.g.*, at 0.8 second). The subsequent migration of air bubbles away from the wall is less significant than that in the single bubble model. This indicates a difference between a single bubble simulation and a continuous bubbling simulation. In the continuous bubbling model, a bath circulation is formed due to drag between the bubble and liquid, and this liquid flow then acts to push air bubbles closer to the anode sidewall.

2. Determination of quasi-stable state

After a period of continuous gas injection, the bath flow will reach a quasi-stable state. It is necessary to know how long the bath will take to reach the quasi-stable state, since only then can time-averaged parameters be processed to compare the two systems.

Normally, from a given initial condition, a flow undergoes an initial transient stage and finally reaches a quasi-stable state regime in which flow variables oscillate around a mean value. Point 1 ($x = 5$ mm, $y = 75$ mm, $z = 50$ mm) and Point 2 ($x = 195$ mm, $y = 75$ mm, $z = 50$ mm) in Figure 2 are used to determine the time to reach the quasi-stable state.

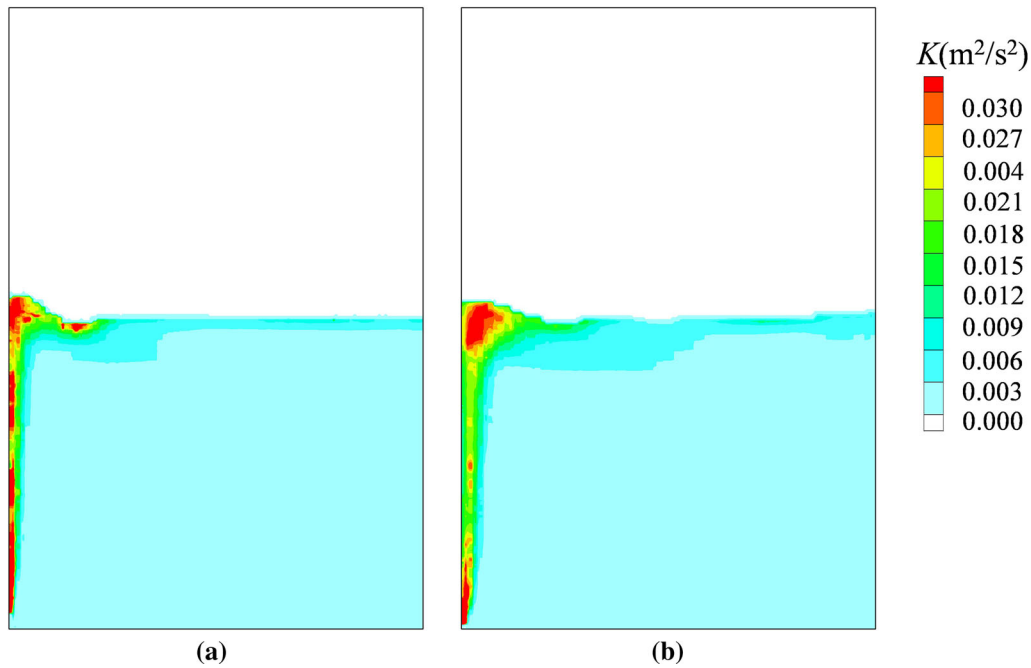


Fig. 15—Spatial distribution of time-averaged turbulence kinetic energy: (a) Case 4, CO₂-cryolite system; (b) Case 5, air-water system.

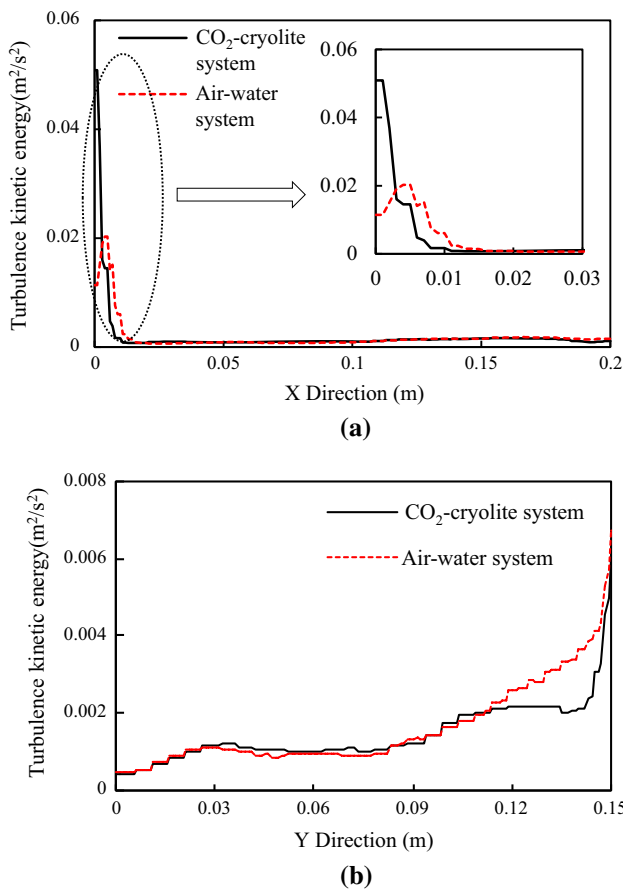


Fig. 16—Time-averaged turbulence kinetic energy on two lines for both cases: (a) $Y = 75$ mm; (b) $X = 100$ mm.

The velocity changes at these points as a function of time are shown in Figure 12. We can see the values of velocity climb from 0 to a quasi-stable value after a period of time and fluctuate around a certain value. Point 1 is above the inlet boundary at half the liquid depth, for which the velocity reaches its stable state relatively early in the simulation (about $t = 0.5$ second) albeit with large rapid fluctuations which continue for the entire simulation. Point 2 is near the outer wall at half the liquid depth, and it requires more time for a quasi-stable value to be established (about $t = 5$ second). We also measured several other points in this domain and their fluctuations show a similar trend to that of Point 1 and Point 2. Thus, the quasi-stable state of the simulation is reached after a flow time of 5 seconds.

In the next section, time-averaged flow characteristics, namely flow patterns, turbulence kinetic energy, turbulence dissipation rate, and gas volume fraction, will be averaged in the period from 5 to 20 seconds.

3. Time-averaged flow characteristics

a. Time-averaged flow patterns. Figure 13 shows the time-averaged flow patterns on the vertical middle plane (see Figure 2 for the location of measuring plane $Z = 50$ mm) of the computing domain. The velocity vectors are plotted for a visual comparison. The most striking common feature of the two systems is the clockwise vortex that is driven by the rising bubbles. The bubbles rise upward along the anode sidewall due to the buoyancy force, which drives liquid upward. The liquid flow changes its direction to horizontal at the liquid surface and flows toward the outer wall. Then the liquid

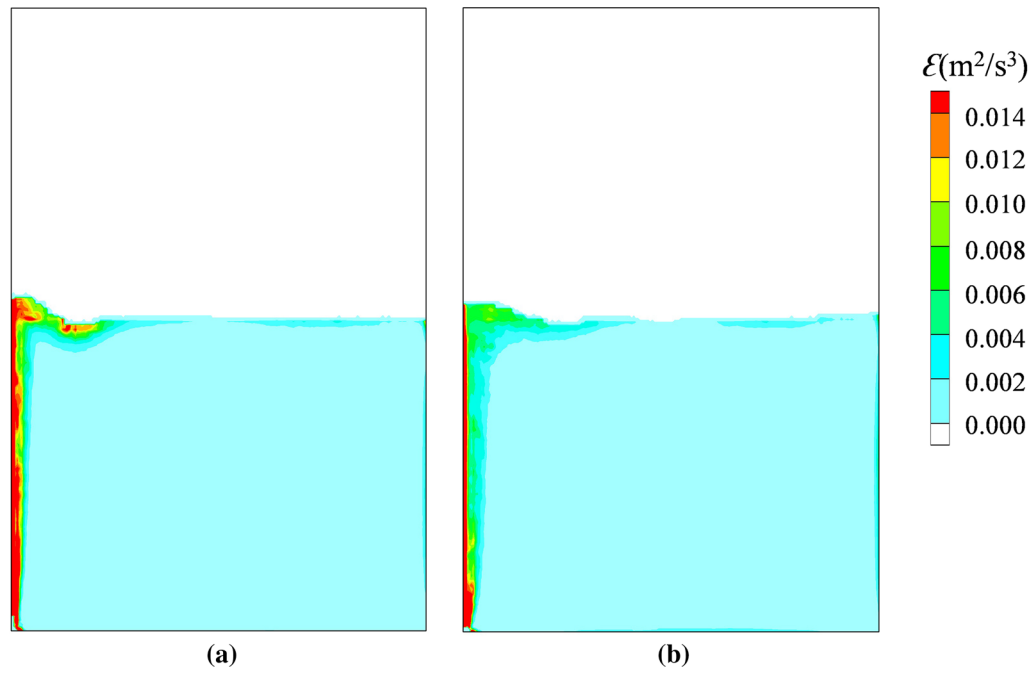


Fig. 17—Spatial distribution of time-averaged turbulence dissipation rate for both cases: (a) Case 4, CO₂-cryolite system; (b) Case 5, air-water system.

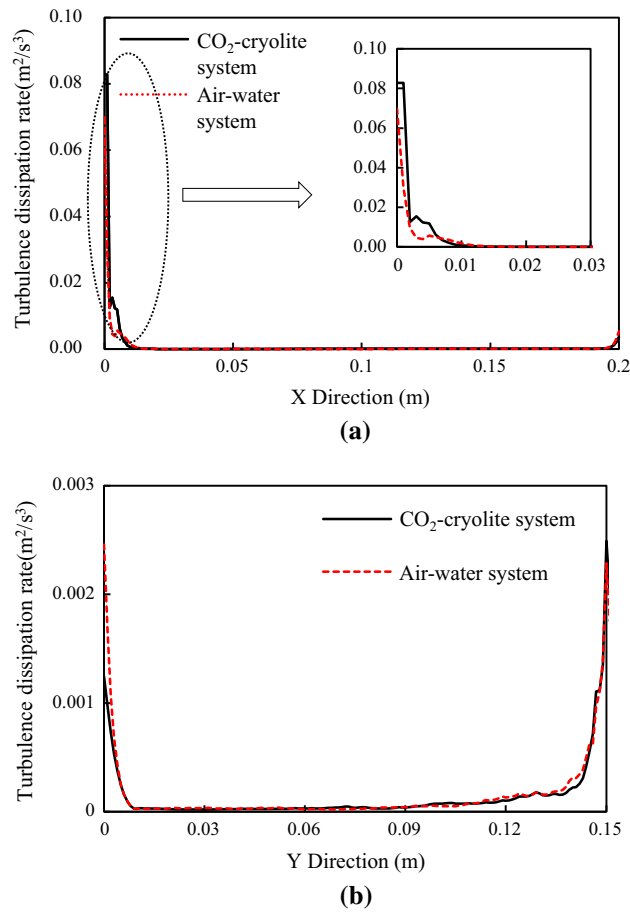


Fig. 18—Time-averaged turbulence dissipation rate at two lines for both cases: (a) $Y = 75$ mm; (b) $X = 100$ mm.

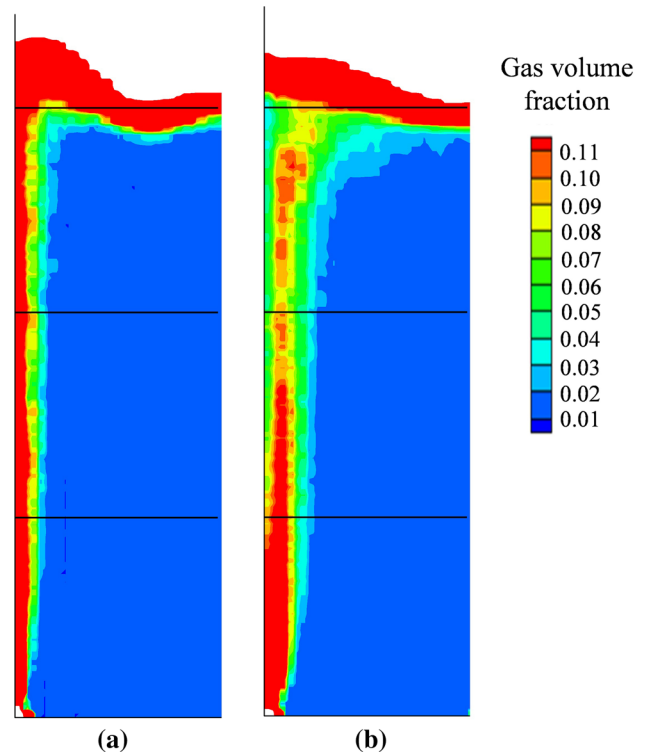


Fig. 19—Spatial distribution of time-averaged gas volume fraction in the bubble plume region. (a) Case 4, CO₂-cryolite system; (b) Case 5, air-water system.

flows downward in the area close to the outer wall, across the bottom of the domain, and rejoins the upward flow near the inlet.

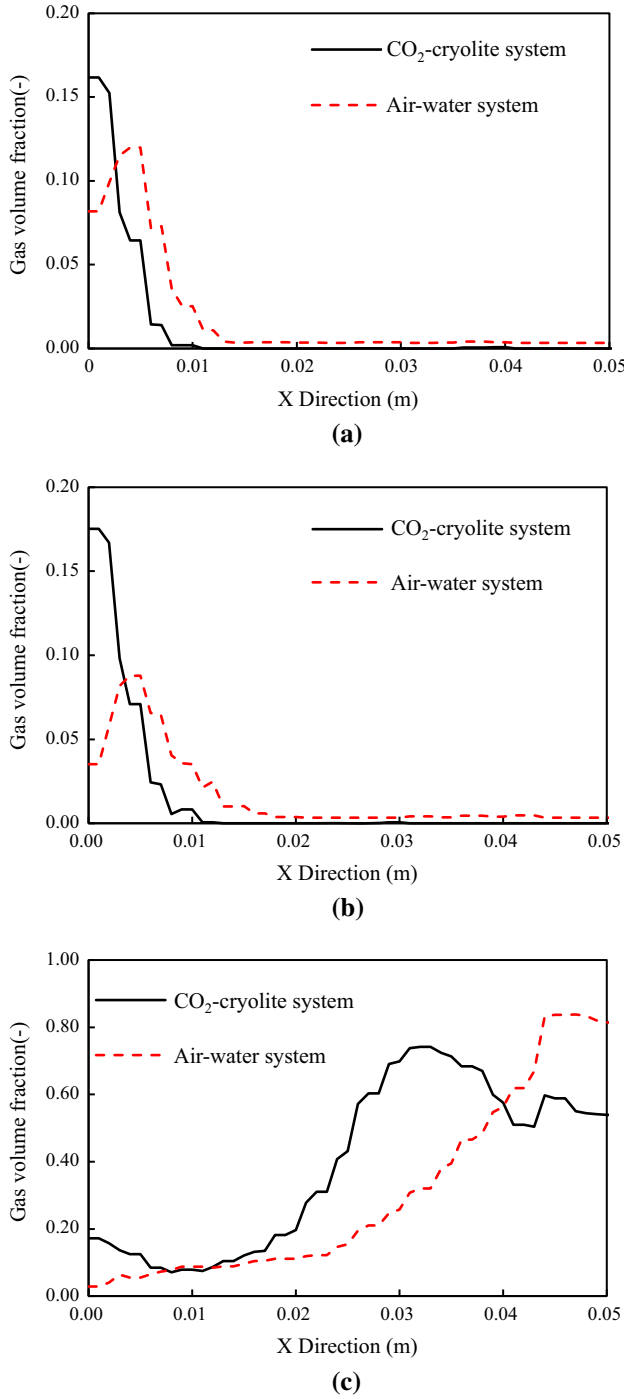


Fig. 20—Time-averaged gas volume fraction at two lines for both cases: (a) $Y = 50$ mm; (b) $Y = 100$ mm; (c) $Y = 150$ mm.

Figure 13 also plots the velocity contours. The value of velocity ranges from 0 to 0.45 m/s for both cases. The largest liquid velocity is located at two areas: (a) the area near the anode sidewall due to the strong buoyancy force on bubbles and (b) the area at the liquid surface, where the upward moving liquid reaches the liquid surface and changes its direction toward the outer wall. Further quantitative comparison between the two systems is given in Figure 14: vertical and horizontal

velocity components on the red line $Y = 75$ mm and the black line $X = 100$ mm are plotted. Only small differences can be seen: the agreement of flow pattern between CO_2 -cryolite system and air-water system is generally good. The main difference is that the maximum velocity is slightly higher for the air-water system.

b. Time-averaged turbulence kinetic energy (K) and turbulence dissipation rate (ε). Bubbles rising in the liquid give rise to increased turbulence of the liquid phase, known as bubble-induced turbulence. Various models have been proposed in the literature to account for this mechanism, with the two most widely accepted being modifying turbulence eddy viscosity by adding a bubble-induced term,^[52,74] and adding a source of bubble induced turbulent kinetic energy. Bubble-induced turbulence is very case dependent and is still an active area of research, so there is not yet agreement on a universal source term for general use.^[74] Most of the published research relate to vertical pipe bubbling flow which will be different from less confined flow. The detailed transient flow offers an excellent opportunity to assess the range of bubble-induced turbulence.

The key variable, turbulence kinetic energy (K), characterizes velocity fluctuations associated with eddies. The turbulence energy dissipation rate (ε) is another important turbulence variable in transient flow. It refers the rate of transformation by dissipation of turbulent kinetic energy to kinetic molecular energy (or heat energy). The time-averaged value of K and ε can be given by the following Eqs. [6] and [7]^[75]:

$$K = \frac{1}{2} \overline{u'_i u'_i} \quad [6]$$

$$\varepsilon = \nu \times \left(\overline{\frac{\partial u_i}{\partial x_k}} \right) \left(\overline{\frac{\partial u_i}{\partial x_k}} \right), \quad [7]$$

where K is turbulence kinetic energy, m^2/s^2 ; ε the turbulence dissipation rate m^2/s^3 ; ν is the kinematic viscosity of fluid, m^2/s ; u' is the turbulent velocity component, m/s ; u is the velocity component, m/s ; and i, j, k denote the Cartesian coordinate directions.

Figure 15 shows resulting contours of K for the two systems. A similar distribution can be seen with the highest K values generally located in the bubble plume region. Liquid splashing at the top liquid surface near the anode sidewall also produces a region of high turbulence. The value of K ranges from 0 to $0.03 \text{ m}^2/\text{s}^2$ in the two cases, which is of the same magnitude as, but a bit higher than the published PIV measured results from a water model (0 to $0.012 \text{ m}^2/\text{s}^2$).^[15] The larger values in the simulation may due to including the three components in Eq. [6], while the experiment only measured two velocity components. These PIV measurements have previously been used to adjust the source term of bubble-induced turbulence in two-phase bath flow models.^[53] The continuous bubbling model would be very helpful to further refine the source term of bubble-induced turbulence.

Figure 16 shows more quantitative comparisons between the two systems based on the two lines ($Y = 75$ mm and $X = 100$ mm) as shown in Figure 13. For the horizontal line, the turbulence level near the anode side wall is higher in the CO_2 -cryolite system than that in the air-water system. This is because the bubble was attached to the anode side wall during its rise for the CO_2 -cryolite system. For the vertical line, the turbulence levels are very similar in the bottom region; at the surface region, the value of air-water model is higher. It appears that the liquid splashing in the air-water system is stronger than that in the CO_2 -cryolite system.

Figure 17 shows the distribution of ε in which the magnitude ranges from 0 to $0.014 \text{ m}^2/\text{s}^3$ for both systems. Figure 18 gives a quantitative comparisons along two lines ($Y = 75$ mm and line $X = 100$ mm), which again shows similar results for the two cases. The distribution of ε is similar to that of K , indicating the close relationship of the two above variables. As no universal correlations exist for the bubble-induced turbulence and turbulence dissipation rate source terms, physical measurements are used to determine appropriate values on a trial-and-error basis. As stated in one study,^[74] when the flow rate was doubled, the value of the fitted coefficient needed to be raised by a factor of ten, which might seem physically unlikely. To the author's knowledge, there is no published paper reporting the detailed distribution of ε in gas-liquid systems. Thus, the importance of this continuous bubbling model is not only to clarify the difference between the two systems, but also to provide an appropriate formula to express bubble-induced turbulence, which will be addressed in future work.

c. Time-averaged gas volume fraction. In an aluminum cell, the bubble coverage under the anode increases cell resistance and rising bubbles near the anode vertical side walls is the main driver for bath flow in the side channels and thus alumina mixing within the bath.^[53] In previous studies on bubble distribution, the focus was on the bubbles under the anode bottom. It is interesting to compare the bubble distribution in the side channels between the two systems as the bubble plume region will affect the bath flow and mixing patterns.

Figure 19 shows the time-averaged gas volume fraction in the vertical middle plane (see Figure 2 for location of measuring plane $Z = 50$ mm). In order to illustrate the distribution more clearly, only the 50-mm-wide region near the anode side is plotted for comparison as there are almost no bubbles in the rest of the side channel. Although bubbles behave slightly differently in their morphologies and wall contact, the gas distribution was predicted to follow a similar trend, *e.g.*, the plume widens toward the surface. The maximum gas fraction lies in the red area, which indicates the bubble frequency is relatively higher than other places; this can be treated as the path of gas stream.

Figure 20 illustrates more quantitative comparisons of the gas volume fraction on the horizontal lines $Y = 50$ mm, $Y = 100$ mm, and $Y = 150$ mm (the black lines in Figure 19). At the lower part of the side

channel, *e.g.*, $Y = 50$ mm and $Y = 100$ mm, the CO_2 volume fraction decreases in the X direction away from the wall, while the air volume fraction increases first and then decreases. This behavior again shows that the effect of contact angle with CO_2 remains attached to the wall while the air becomes detached and bubbles moves away from the wall. At the top gas-liquid interface, gas volume fraction shows a totally different trend to that lower in the bath, the air and CO_2 volume fractions first increase and then decrease. The phenomenon closely relates to the distribution of the gas stream and bubble behavior as it breaks through and causes slashing at the free surface. In addition to clarifying the difference between the two systems, the gas distribution provides valuable data to adjust the bubble-induced turbulence dispersion force so as to improve quantitative predictions of bath flow at a full cell scale.^[53]

IV. CONCLUSION

This paper reports a further numerical comparison between air-water and CO_2 -cryolite systems using a 3D VOF model for both single bubble and continuous bubbling situations.

The main conclusions from the single bubble model are

1. The CO_2 bubble sliding under an anode in cryolite has a smaller bubble thickness, a larger aspect ratio, and a higher sliding velocity than that of an air bubble in water.
2. Bubbles detach from the side wall as they rise in the side channel in the air-water system, but they remain attached to the wall in the cryolite system.
3. Through dimensionless analysis and numerical simulation, it is concluded that the contact angle is the dominant factor producing these differences; the effects of kinematic viscosity, surface tension and density are very small.
4. Previous 2D simulations showed a similar trend to the present 3D simulation in terms of the difference between the two systems; however, there are additional differences in terms of bubble shapes and bubble thickness that are predicted by the 3D model. 3D simulation is necessary for quantitative studies.

The major conclusions from the continuous bubbling simulations are

1. Both systems show similar bubble morphologies: the gas is dispersed into various sized bubbles after gas turns upward at the anode corner, and the most bubbles are initially attached to the anode sidewall. As in the single bubble simulations, air bubbles detach from the anode sidewall in the air-water system, but they migrate less rapidly away from the wall following detachment than in the single bubble simulation.
2. The time-averaged flow pattern shows a similar trend for the two systems: a recirculation zone is formed in the side channel with an upward liquid flow close to the anode and a downward flow near the side wall.

The value of velocity ranges from 0 to 0.45 m/s for both cases, and the highest values are located in the same areas in the two cases: near the anode vertical wall and the liquid surface.

3. Time-averaged K and ε values show that they are closely related with the trajectories of gas movement. Values of K are in the range of 0–0.03 m²/s², similar to experimentally measured values, and ε values are in the range of 0 to 0.014 m²/s³. The K distribution is very similar to the ε distribution, and when comparing the two systems, they are also very similar in terms of distribution and magnitude.
4. The time-averaged gas volume fraction in the side channel appears to follow a similar trend for both systems: the bubble plume region widens toward the surface, and values of the bubble volume fraction are in a similar range.

From the current numerical comparisons, it is believed that the air–water model is a close substitutive model to study bubble-driven bath flow in aluminum smelting process. However, because of the difference in bubble morphologies between the two systems, and also the reactive generation and growth of bubbles in the real system, there will likely be some differences in bubble coverage of the anode in the anode–cathode gap.

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NOMENCLATURE

i	Anode current density, A/cm ²
T	Absolute temperature, K
P	Pressure, Pa
q	Gas generation rate at the unit anode bottom surface, m ³ /(m ² s)
a	The length of anode bottom surface, mm
b	The width of anode bottom surface, mm
l	The length of inlet boundary, mm
w	The width of inlet boundary, mm
V	Gas injection velocity at the inlet boundary, m/s
ρ_l	Liquid density, kg/m ³
ρ_g	Gas density, kg/m ³
$\Delta\rho$	Density difference between gas and liquid
μ_l	Liquid dynamic viscosity, Pa s
σ_l	Liquid surface tension, N/s
d_g	Equivalent diameter of bubble, mm
v_g	Bubble velocity, m/s
t	Time, s
L	The length of sliding bubble, mm
W	The width of sliding bubble, mm
K	Turbulence kinetic energy, m ² /s ²
u'	Turbulent velocity component, m/s

u	Velocity component, m/s
R	Universal gas constant, 8.314 J/(mol K)
F	Faraday's constant, 96486 A/(mol s)
g	Gravitational acceleration, 9.81 m/s ²

GREEK LETTERS

θ	The contact angle
ε	The turbulence dissipation rate m ² /s ³
ν	The kinematic viscosity, m ² /s

SUBSCRIPTS

i, j, k	Subscripts denoting Cartesian coordinate directions
g, l	Subscripts denoting gas and liquid

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